

ARYAN AGARWAL

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EXPERIENCE

Acorai AB, R&D Engineer

October 2023 – Present

- Created a signal processing pipeline in 3 months for an ECG, PCG, SCG, and PPG wearable cardiovascular sensor in Python that refined physiological time-series sensor data for signal quality assessment
- Modeled early-prototype in SolidWorks with 25+ working components optimized for low-noise data acquisition according to design specifications, requirements, constrained form factors, and manufacturability in 3 months
- Executed 20+ benchtop and on-body test cases on-site across 100+ hardware configurations and diverse user populations to optimize sensor performance by 50% in 4 weeks
- Performed ISO 13485-compliant design controls in collaboration with regulatory team by defining 20+ system and technical requirements, generating V&V test cases and calibration protocols, and conducting risk analysis
- Collaborated with vendors to integrate off-the-shelf and industrial equipment with 5 CAD-designed benchtop test platforms to characterize sensor noise, repeatability, and performance benchmarks on 400+ sensor measurements
- Built a scalable Python data analytic pipeline that generated 200+ page reports containing statistical summaries, pass/fail evaluations, and plots evaluating sensor performance across 80+ sensor KPIs
- Documented 50+ test protocols, study outcomes, product decision processes, and design history files according to ISO 9001 in Jira and Qualio to support FDA regulatory submission
- Validated device compliance with medical and EMC standards (IEC 60601-2-25, ISO 14117) in collaboration with contract manufacturers and regulatory teams
- Presented root-cause analysis identifying key sensor performance issues to 6+ industry expert advisors, translating complex technical data into actionable insights

forMED Technologies, Lead Engineer

June 2021 – October 2023

- Managed 5 engineers in a fast-paced, early-stage startup developing experimental optical system for product design and testing of a non-invasive IOP monitoring system using novel laser imaging technology
- Designed and manufactured portable optical prototype hardware to image eye with motorized movements of 6+ components using SolidWorks, 3D printing, and off-the-shelf parts in 1 month
- Developed and validated board mount digital sensor to process and report real-time eye pressure signals
- Presented experimental results of novel technology to panels of 10+ industry and academic experts

Blacklight Surgical, Software Intern

June 2021 – September 2021

- Optimized SVM model by 20% to classify cancer tissue by testing combinations of 42 training data parameters
- Created a Python algorithm to analyze electrical performance of multi-channel fluorescence spectroscopic system

King Lab, Research Lead

May 2020 – May 2022

- Led cross-functional team of 5 engineers in designing wearable sensor for real-time body temperature analysis
- Programmed and tested Android application using Java for wireless BLE connection, signal processing, statistical analysis, and informed medical data visualization on custom UI in 6 months

EDUCATION

University of California, San Diego

December 2023

- Master of Science in Electrical and Computer Engineering
- Degree Focus in Medical Devices and Systems

University of California, Irvine

June 2022

- Bachelor of Science in Biomedical Engineering, *Magna Cum Laude*

SKILLS

Software: Matlab, Python, LabView, Java, Git, Bitbucket, Jupyter, Android Studio, C++, SolidWorks, ImageJ, Microsoft Office, Zemax, CrunchIt, Machine Learning, Computer Vision, Jira, Confluence, Qualio

Engineering: Design Controls, Rapid Prototyping, Signal and Image Processing, System Design and Analysis, 3D printing, Test Protocols, Technical Writing, Oral Communication, Manufacturing, Laboratory Instrumentation